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EECE Department

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Assignment 2

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1 Introduction

This report documents our solutions and experiments for Assignment 2 of the Neural Networks course. The assignment is divided into two primary parts:

- **Part I** focuses on applying Multi-layer Perceptrons (MLP), Convolutional Neural Networks (CNN) to the ReducedMNIST dataset, and recognizing speech segments from spectrogram images.
- **Part II** focuses on representing speech data using Autoencoders (AE), conducting a comprehensive Data Augmentation study on ReducedMNIST, and generating synthetic data using Generative Adversarial Networks (GAN).

Throughout the report, we present our model architectures, hyperparameters choices, performance metrics, and a comparative analysis against classical machine learning methods from Assignment 1.

2 Part I: Neural Networks on Image and Audio Data

2.1 Problem 1: Multi-layer Perceptron (MLP) on ReducedMNIST

In this problem, we designed an MLP model using the ReducedMNIST dataset (1000 examples per digit for training, 200 per digit for testing). We trained variations of the MLP including models with 1, 3, and 4 hidden layers, using the DCT features calculated in Assignment 1.

2.1.1 Methodology

2.1.2 Results and Observations

2.2 Problem 2: Convolutional Neural Network (CNN) on ReducedMNIST

In this problem, we switch to working with the raw 28×28 image pixels to train a Convolutional Neural Network. We adapted a LeNet-5 style architecture with ReLU activations to suit the ReducedMNIST dataset characteristics without explicit manual feature extraction.

2.2.1 Baseline Architecture

The baseline CNN directly takes image pixels through two convolutional layers (with 5×5 kernels) followed by Maximum Pooling, and then propagates features through three fully connected layers to output class predictions.

2.2.2 Hyperparameter Variations

To explore model performance dynamically, we made the following major variations to the baseline model:

- **Variation 1 (Wider):** Increased the number of convolutional filters and the width of the fully connected nodes to provide excessive capacity.
- **Variation 2 (LeakyReLU):** Swapped the ReLU activation functions out for LeakyReLU models to check the impact of non-zero gradient limits on inactive regions.
- **Variation 3 (Dropout):** Added Dropout probabilities to fully connected layers to test its regularization effect.

2.2.3 Results and Comparison

Table 1 illustrates the accuracy and computing times for the tested configurations.

Table 1: CNN Variation Results for Problem 2

Variation	Description	Acc. (%)	Train (ms)	Test (ms)
BaseCNN	Baseline LeNet-style CNN with ReLU	95.0	74032.0	11554.2
Variation 1	Increased filters and FC widths	97.3	18493.0	521.1
Variation 2	ReLU replaced by LeakyReLU	96.2	16089.9	490.3
Variation 3	Added dropout before FC2/FC3	96.7	16424.0	453.8

Discussion

2.3 Problem 3: Speech Recognition from Speech Spectrum

This problem extends our models to audio classification by using spectrograms arrayed from audio waveforms representing 10 digits spoken by multiple people.

2.3.1 Baseline CNN Training

2.3.2 Data Augmentation (Speech-domain)

2.3.3 Data Augmentation (Spectrum-domain)

2.3.4 Combined Augmentations

3 Part II: Data Augmentation and Synthesis

3.1 Problem 4: Autoencoder for Utterance Representation

This problem demonstrates how to compress variable-length sequence data (specifically 15 millisecond audio frames concatenated) into fixed-size representation using an Autoencoder (AE).

3.1.1 Methodology

3.1.2 Results

3.2 Problem 5: Data Augmentation on ReducedMNIST

In this problem, we systematically study the effect of artificially expanding the dataset size over an underlying algorithm using augmentation parameters natively applied (Delta shifts, rotations, noise).

3.2.1 Test Mix Analysis

Table 2: Data Augmentation performance across various data size regimes

Augmented (# per digit)	**350 Real**	**750 Real**	**1,000 Real**
1,000 Generated	-	-	-
1,500 Generated	-	-	-
2,000 Generated	-	-	-

Discussion

3.3 Problem 6: Generative Adversarial Networks (GAN)

This section expands upon our dataset generation from the augmentation strategy to true generation techniques. A GAN was used to model the ReducedMNIST dataset using 350 training examples correctly.

3.3.1 Visual Analysis of Generated Digits

3.3.2 Testing Pipeline on Synthesized Data

3.3.3 Comparing Results

4 Appendices

.1 Code Snippets